# ContextWorld: Benchmarking In-Context Dynamics Adaptation in Latent Visual World Models


## Abstract

*Latent world models learn environment dynamics to predict future states from visual observations and action sequences. These dynamics are not always fixed, and interaction history can help models adapt their predictions to different conditions. Yet low prediction error does not necessarily indicate that models have learned to adapt from interaction history. We introduce ContextWorld, a benchmark for evaluating in-context dynamics adaptation in action-conditioned latent world models without weight updates. ContextWorld uses paired counterfactual evaluations that vary interaction histories while keeping the current observation and future actions fixed. This design measures both whether predictions distinguish the correct future and whether they benefit from the corresponding history. The benchmark spans nine tasks across four environments, with separate training and evaluation data. Evaluating three latent world models, LeWM, PLDM, and DINO-WM, reveals that models trained under fixed dynamics make little use of interaction history on most tasks. Training with multiple dynamics can improve adaptation, but the effectiveness of data scaling and training strategies varies substantially across tasks. We further find that training loss can decrease while models still fail to capture dynamics from interaction history, even on the training distribution. These findings expose a gap between optimizing prediction loss and in-context dynamics adaptation, highlighting the need to evaluate whether models use interaction history to adapt their predictions.*


## 1. Introduction

Visual world models learn environment dynamics to predict future states [19], with latent world models performing these predictions within learned representations of visual observations [20, 21]. A model trained under fixed dynamics can achieve high prediction accuracy without utilizing interaction history to adapt its predictions to current conditions. However, this does not necessarily mean that the model can adapt to changes in environment dynamics [33]. For example, visually identical objects can move different distances under the same push if surface friction differs. Interaction history provides evidence about these hidden physical properties. We investigate whether latent world models can leverage this contextual evidence to predict future states under varied dynamics *without updating their weights*—an ability we term *in-context dynamics adaptation*.

Existing physical prediction benchmarks have examined how interaction history supports reasoning about hidden properties and future outcomes [4, 32]. Evaluating in-context dynamics adaptation in latent world models requires examining not only whether predictions distinguish different dynamics, but also whether they benefit from the corresponding interaction history. We introduce ContextWorld (Figure 1) to evaluate these complementary aspects directly through action-conditioned latent predictions. The benchmark spans nine tasks across four environments, with separate training and evaluation data.

In its paired tasks, two dynamics conditions share the same current observation and future action sequence, but produce diverging ground-truth futures. Because a deterministic model that ignores history receives identical inputs and yields identical predictions for both conditions, it scores at most 50% on paired future selection. This paired counterfactual design restricts history-agnostic heuristics, enabling us to concurrently evaluate two aspects directly in latent space: (1) adaptation accuracy, which measures whether a model's prediction is closer to its corresponding true future than to the alternative condition's future; and (2) history sensitivity, which measures whether matching history yields lower prediction error than mismatched history.

Using ContextWorld, we benchmark three representative latent world models: LeWM [24], PLDM [30], and DINO-WM [37]. We find that models trained under fixed dynamics make negligible use of interaction history across most tasks. To examine how this adaptation ability can be acquired, we evaluate the effects of multi-dynamics training, data scaling, and training strategies. Our experiments reveal that the effectiveness of these training choices varies substantially across tasks. We further find that training loss can decrease while models still fail to use interaction history to distinguish dynamics conditions, even on the training dis-

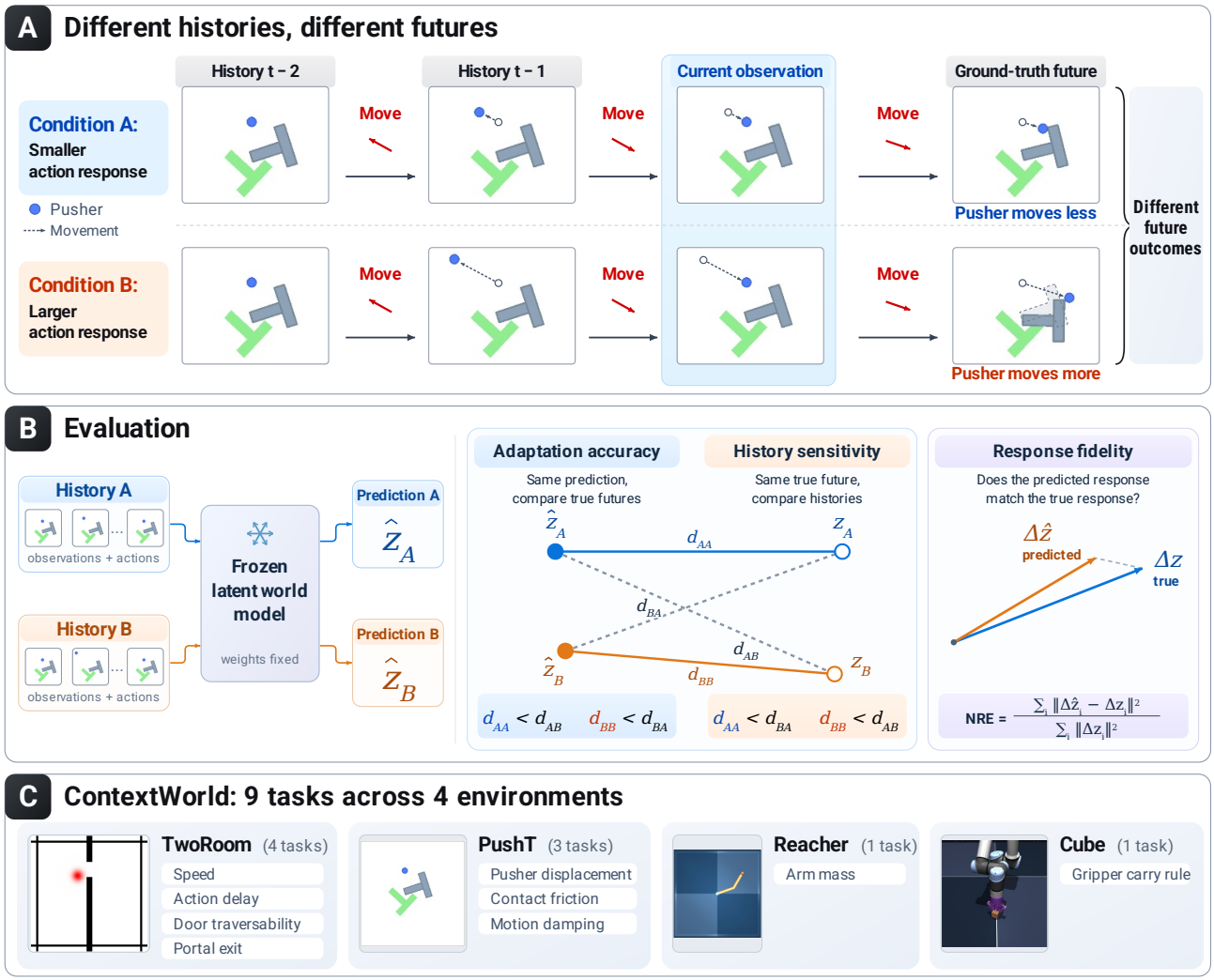



Figure 1. **ContextWorld overview.** (A) A recorded pusher-displacement pair with shared current observations and future actions. Red arrows show action directions; dashed arrows mark observed endpoint displacement. The translucent outline in the lower future panel marks the T block's current pose. These annotations are not model inputs; all scene images use the same crop of the original observations. (B) The frozen model produces latent predictions from each history. Three complementary checks evaluate paired tasks. Filled circles denote predictions and open circles denote encoded true futures. Accuracy and history sensitivity share one four-point diagram: accuracy compares true futures for a fixed prediction, whereas history sensitivity compares histories for a fixed true future. Here $d_{ij}$ is the error between the prediction from history $i$ and true future $j$; the inequalities show the desired ordering for both conditions. Response fidelity compares $\Delta\hat{z} = \hat{z}_B - \hat{z}_A$ with $\Delta z = z_B - z_A$, using the same schematic points. All latent-space geometry is illustrative, not a projection of measured model outputs. (C) Nine tasks across four environments.

tribution. This reveals a gap between optimizing prediction loss and in-context dynamics adaptation.

Our main contributions are:

- **ContextWorld benchmark.** We introduce nine tasks involving hidden dynamics across four environments, with separate training and evaluation data. The benchmark includes paired counterfactual evaluations that measure adaptation accuracy and history sensitivity directly from action-conditioned latent predictions.
- **Systematic evaluation.** We benchmark three representative latent world models across data scales and training strategies, revealing substantial differences in adaptation across tasks. We further show that training loss can decrease even when models fail to leverage interaction history on the training distribution.

## 2. Related Work

**Visual world models.** Visual world models learn environment dynamics to simulate and predict future states. Early frameworks [19] combined learned visual representations with recurrent dynamics to support internal policy training. Subsequent methods advanced this paradigm through online planning (PlaNet [20]) and policy learning via imagined latent trajectories (Dreamer [21]). While these early approaches rely on pixel-level image reconstruction to learn representations, Joint-Embedding Predictive Architectures (JEPAs) [23] shift the training objective toward predicting target representations directly without reconstructing pixels. This principle has been applied to masked spatial regions (I-JEPA [2]) and extended to temporal video representation learning (V-JEPA [5]). To support robotic planning, V-JEPA 2 [3] introduces an action-conditioned predictor trained on robot interactions atop pretrained video representations.

Action-conditioned latent world models also differ in how they obtain their visual representations. DINO-WM [37] trains a predictive dynamics model on frozen pretrained features for planning from offline trajectories. In contrast, PLDM [30] jointly optimizes the visual encoder and the predictor using reward-free offline data. To stabilize joint training, LeWM [24] simplifies the objective to a latent prediction loss and a single representation regularizer. We benchmark these three paradigms to evaluate how structural choices influence in-context dynamics adaptation.

**In-context adaptation.** When environment dynamics differ between training and deployment, interaction history provides a signal for adaptation without requiring weight updates. Meta-reinforcement learning [6] leveraged this principle by training agents to adapt from limited experience across tasks. $RL^2$ [14] implicitly stores interaction history within a recurrent network's hidden state, while PEARL [29] infers a task representation from experience to condition the policy. Paralleling large language models, which learn from in-context examples without parameter updates [8, 13], this adaptation paradigm has been extended to visual world models.

Within world models, context guides predictions of action outcomes in new environments. OSWM [16] pretrains a transformer on synthetic dynamics, using observed state transitions as context to forecast futures. L2World [33] distinguishes using context to identify previously seen environments from learning new dynamics. It studies the effects of context length and training environment diversity through multi-step prediction errors and context perturbations. For robotic control, ICWM [34] employs task-agnostic interaction history to adapt policies to unseen camera viewpoints without weight updates, using ablations over correct, absent, and mismatched contexts to validate history reliance.

**Benchmarks.** To evaluate adaptation across environments, existing benchmarks introduce controlled variations in dynamics, appearance, or task rules. Meta-World [35] and Procgen [11] test generalization across manipulation tasks and procedurally generated levels. CausalWorld [1] and CARL [7] vary environment properties, while the XLand series (XLand-MiniGrid [26], XLand-100B [27]) evaluates policy adaptation from interaction history across distributions of tasks.

For physical prediction, several benchmarks focus on counterfactual reasoning and hidden property inference. CoPhy [4] asks models to predict outcomes after changes to a scene's initial configuration, using the original trajectory as evidence of hidden properties. Filtered-CoPhy [22] ensures the observed trajectory contains sufficient information for the counterfactual prediction. ComPhy [9] probes reasoning regarding hidden mass and charge through predictive and counterfactual video querying. Physion++ [32] tests whether interaction history helps predict future contact in paired trials that share the same current observation but differ in physical properties, assessing this understanding via auxiliary readout classifiers.

Beyond visual domains, a recent time-series benchmark [36] reveals that low prediction error can coexist with incorrect response directions when actions change. This echoes our finding of a mismatch between prediction-loss minimization and true dynamics adaptation. A remaining question is when action-conditioned latent world models learn to leverage history for accurate prediction. ContextWorld addresses this gap through controlled comparisons across models, tasks, data scales, and training strategies.

## 3. ContextWorld Benchmark

ContextWorld is designed to evaluate in-context dynamics adaptation—specifically, whether models use interaction history to accurately predict future states conditioned on an action sequence. This section formalizes the prediction setting, describes how we construct the tasks and datasets, and explains the paired evaluation protocol and metrics.

### 3.1. Problem Formulation

To isolate in-context adaptation from weight updates, model parameters $\theta$ remain fixed during evaluation. The model receives exclusively visual observations and actions. For a target horizon $\tau$, the model receives an interaction history $H_t$, comprising interleaved visual observations up to time $t$ and their executed actions, alongside a future action se-

Table 1. Overview of the nine tasks evaluating in-context dynamics adaptation across four environments. History length $h$ counts context observations (including the current observation) provided to the model. Sampling details and simulation settings appear in Appendices A and D.

| Task | Environment | History length $h$ | What history helps the model predict |
|---|---|---|---|
| Speed | TwoRoom | 3 | How far the agent moves under the same action |
| Pusher displacement | PushT | 3 | Whether the same action moves the pusher a shorter or longer distance |
| Arm mass | Reacher | 3 | How the same torque changes arm motion |
| Action delay | TwoRoom | 7 | When an action takes effect |
| Contact friction | PushT | 3 | How friction changes object motion during contact |
| Motion damping | PushT | 3 | How quickly object motion decays after contact ends |
| Gripper carry rule | Cube | 3 | Whether a closed gripper carries the cube |
| Door traversability | TwoRoom | 3 | Whether a visually identical door permits passage |
| Portal exit | TwoRoom | 3 | Where the agent emerges after entering the same portal |

quence $A_{t:t+\tau-1}$. The prediction and its target are

$$\begin{aligned} \hat{z}_{t+\tau} &= f_\theta(H_t, A_{t:t+\tau-1}), \\ z_{t+\tau} &= e_\theta(o_{t+\tau}). \end{aligned} \tag{1}$$

Here, $o_{t+\tau}$ is the ground-truth observation at time $t+\tau$, and $e_\theta$ is the model's target encoder, which is also frozen. Predictions are evaluated directly against these encoded targets in latent space.

### 3.2. Task Design

We construct nine tasks across four environments by varying physical parameters and interaction rules. These environments are TwoRoom navigation [37], PushT [10], Reacher from the DeepMind Control Suite [31], and Cube from OGBench [28]. Figure 1 illustrates the task design, and Table 1 summarizes the tasks and their history lengths. Models are trained separately for each task, and DINO-WM keeps its pretrained visual encoder frozen.

Our main design is to test whether models can use interaction history to predict different outcomes when the current observation and future actions alone cannot distinguish the underlying dynamics. For six tasks—pusher displacement, arm mass, contact friction, motion damping, gripper carry rule, and portal exit—we construct pairs of samples that share the same current observation and future action sequence but have different interaction histories and ground-truth futures. Each sample is generated through a continuous simulation under fixed dynamics. We check relevant physical states, such as velocity and contact state, at the prediction point, since identical images do not necessarily imply identical physical states. For tasks where the two conditions cannot naturally share a starting position, we exchange their starting positions to prevent the initial images from revealing the dynamics condition. A model that ignores history therefore cannot distinguish the two futures within a pair (Appendix B).

The other three tasks examine history-based prediction through their respective task protocols. For Speed, we compare predictions conditioned on histories generated at different speeds, while keeping the current observation and query actions fixed. The model must achieve lower prediction error with the history corresponding to the target speed than with any alternative history. Evaluation includes seen speeds, interpolation, and extrapolation, while the main results use unseen interpolation speeds. For Action Delay, the model receives seven observation frames to infer action delays ranging from 0 to 10 simulator steps. Since the main prediction horizon is five steps, delays of 5–10 produce the same response within this horizon and are grouped into one of six equally weighted response groups (Appendix C). For Door traversability, the model uses preceding interactions to predict whether a door is passable or blocked. Both rules appear during training, while Test uses 42 previously unseen door positions to evaluate generalization to new positions.

### 3.3. Dataset Construction and Splits

To evaluate models on unseen interaction rollouts, all nine tasks provide trajectory-disjoint Training, Development, and Test splits. Crucially, the Test set is reserved for final evaluation, while the Development set supports method and checkpoint selection. Across the six paired tasks, Action Delay, and Door Traversability, models are evaluated on unseen trajectories under dynamics conditions or rules represented in training. In contrast, the speed task assesses parameter generalization, with the main Test results reported for unseen speeds held out within the training range. Each paired task provides up to 32,768 training pairs, alongside 256 balanced counterfactual pairs in each evaluation split (Table 4, Appendix D).

### 3.4. Evaluation Protocol and Metrics

Prediction error alone does not establish in-context dynamics adaptation: models may reduce errors shared across dynamics conditions without predicting their differences. Conversely, selecting the correct future does not guarantee an accurate conditional response: Appendix F shows that a small predicted difference can still yield perfect future-selection accuracy. We therefore evaluate two aspects: Adaptation Accuracy and History Sensitivity. We additionally report Normalized Response Error (NRE) and Multi-Step Prediction Error ($E/B$) to assess conditional response and complete rollout errors.

**Adaptation Accuracy ($A_{\text{future}}$).** For each of the six paired tasks, let $i \in \{1, \dots, N\}$ index the $N$ counterfactual pairs, with dynamics conditions $A$ and $B$. Let $z_i^A, z_i^B$ be the encoded ground-truth futures and $\hat{z}_i^A, \hat{z}_i^B$ the predictions from their respective histories. Write $d(x, y)$ for the mean squared error between latent representations $x$ and $y$. A prediction is correct if it is strictly closer to the future under its own condition than to the alternative:

$$\begin{aligned} I_i^A &= \mathbf{1}[d(\hat{z}_i^A, z_i^A) < d(\hat{z}_i^A, z_i^B)], \\ I_i^B &= \mathbf{1}[d(\hat{z}_i^B, z_i^B) < d(\hat{z}_i^B, z_i^A)]. \end{aligned} \tag{2}$$

Here $\mathbf{1}[\cdot]$ equals one if its condition holds and zero otherwise. The percentage accuracy is

$$A_{\text{future}} = \frac{100}{2N} \sum_{i=1}^{N} (I_i^A + I_i^B). \tag{3}$$

Because the current observation and future actions are shared, a deterministic model ignoring history makes identical predictions for the two conditions and scores at most 50%; ties count as failures. Speed, action delay, and door traversability follow their own scoring rules (Appendix E).

**History Sensitivity (History win rate).** We test whether matching history reduces prediction error by keeping the current observation, future actions, and true future fixed while swapping the history. For paired tasks, define

$$\begin{aligned} J_i^A &= \mathbf{1}[d(\hat{z}_i^A, z_i^A) < d(\hat{z}_i^B, z_i^A)], \\ J_i^B &= \mathbf{1}[d(\hat{z}_i^B, z_i^B) < d(\hat{z}_i^A, z_i^B)]. \end{aligned} \tag{4}$$

The History win rate in percent is

$$A_{\text{history}} = \frac{100}{2N} \sum_{i=1}^{N} (J_i^A + J_i^B). \tag{5}$$

Unlike adaptation accuracy, which compares two targets for a fixed prediction, History win rate compares predictions from two histories against a fixed target; ties count as failures. The other tasks use their corresponding history comparisons (Appendix E).

**Normalized Response Error (NRE).** To assess whether predicted differences between conditions match the true differences, define $\Delta z_i = z_i^B - z_i^A$ and $\Delta \hat{z}_i = \hat{z}_i^B - \hat{z}_i^A$. We compute

$$\text{NRE} = \frac{\sum_i \|\Delta \hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}, \tag{6}$$

where $\|\cdot\|$ is the Euclidean norm in latent space and the denominator must be positive. An NRE of 0 means the predicted difference is exact; a zero predicted difference gives NRE equal to 1. Differences cancel errors shared across conditions, so low NRE does not guarantee accurate complete predictions. Multi-condition tasks use their task-specific response comparisons.

**Multi-Step Prediction Error ($E/B$).** We evaluate open-loop autoregressive predictions at simulator steps 5, 10, 15, 20, and 25, without intermediate ground-truth observations. Let $q$ index evaluation scenes; $k$, dynamics conditions; $c$, shared future action sequences; and $t$, the five evaluation times. Scene $q$ contains $K_q$ conditions and $C_q$ action sequences, with $T = 5$ times. Let $\hat{z}_{qkct}$ and $z_{qkct}$ denote predicted and true latent states, and let $\bar{z}_{qct} = K_q^{-1} \sum_k z_{qkct}$ be the mean true future across conditions for the same scene, actions, and time. The scene-level prediction error $L_q$ and mean-future reference error $B_q$ are

$$\begin{aligned} L_q &= \frac{1}{K_q C_q T} \sum_{k,c,t} \|\hat{z}_{qkct} - z_{qkct}\|^2, \\ B_q &= \frac{1}{K_q C_q T} \sum_{k,c,t} \|z_{qkct} - \bar{z}_{qct}\|^2. \end{aligned} \tag{7}$$

We report their aggregate ratio [25]:

$$E/B = \frac{\sum_q L_q}{\sum_q B_q}, \tag{8}$$

Table 2. Training schemes. Mixed combines original-environment and ContextWorld training windows at a 1 : 1 sampling ratio. Original → Mixed starts from an Original checkpoint, with Joint updating the visual encoder and Frozen Encoder keeping it fixed. DINO-WM uses a frozen pretrained visual encoder in all schemes.

| Scheme | Start | Data | Encoder |
|---|---|---|---|
| Original | Initial | Original | Updated† |
| Mixed | Initial | Mixed | Updated† |
| Original → Mixed (Joint) | Original | Mixed | Updated |
| Original → Mixed (Frozen Encoder) | Original | Mixed | Frozen |

Initial weights are random for LeWM and PLDM. DINO-WM uses a pretrained visual encoder and a random predictor. †DINO-WM keeps its encoder frozen in every scheme; its continued-training scheme is labeled Original → Mixed. Other modules remain trainable; native training losses are retained.

Table 3. **Adaptation metrics on the Test split.** Each cell lists Accuracy (%) ↑, History win rate (%) ↑, and NRE ↓ from top to bottom. Entries are means; run counts and standard deviations appear in Table 6. Speed uses the same scoring rule for Accuracy and History win rate. Bold marks the highest Accuracy and History win rate and lowest NRE per task, including ties at the displayed precision. NRE is measured in each checkpoint's target representation. A dash denotes no comparable checkpoint.

| Model | Training scheme | Speed | Pusher displacement | Arm mass | Action delay | Contact friction | Motion damping | Gripper carry rule | Door traversability | Portal exit |
|---|---|---|---|---|---|---|---|---|---|---|
| LeWM | Original | 58.04<br>58.04<br>0.896 | 48.96<br>28.32<br>1.018 | 50.07<br>51.63<br>0.977 | — | 49.93<br>50.13<br>1.001 | 50.00<br>48.63<br>1.007 | 52.21<br>73.76<br>0.909 | 50.33<br>50.94<br>0.996 | 50.07<br>51.95<br>0.990 |
| | Mixed | 98.19<br>98.19<br>0.042 | 50.78<br>79.10<br>0.974 | 85.74<br>89.84<br>0.045 | 97.70<br>99.18<br>0.013 | 50.00<br>49.41<br>1.000 | 50.00<br>48.24<br>1.004 | 50.00<br>49.80<br>1.000 | 99.83<br>**100.00**<br>0.041 | 50.78<br>76.37<br>0.922 |
| | Original → Mixed (Joint) | 96.89<br>96.89<br>0.055 | 91.80<br>99.02<br>0.215 | 87.30<br>93.36<br>0.038 | — | 50.39<br>54.30<br>0.999 | 50.00<br>49.61<br>1.001 | 50.00<br>52.34<br>0.997 | 99.83<br>**100.00**<br>0.038 | 50.39<br>66.80<br>0.934 |
| | Original → Mixed (Frozen Encoder) | **98.44**<br>**98.44**<br>**0.030** | 97.46<br>98.83<br>0.128 | 77.15<br>86.52<br>0.346 | — | **86.13**<br>**92.77**<br>**0.429** | 53.91<br>61.91<br>0.958 | **76.37**<br>**84.57**<br>**0.271** | **100.00**<br>**100.00**<br>0.036 | 77.93<br>94.34<br>0.264 |
| PLDM | Original | 48.41<br>48.41<br>0.922 | 45.57<br>9.38<br>1.058 | 51.82<br>66.99<br>0.921 | — | 50.00<br>49.09<br>1.014 | 50.26<br>50.52<br>0.947 | 50.78<br>57.42<br>0.937 | 50.06<br>51.50<br>0.997 | 50.07<br>51.50<br>0.996 |
| | Mixed | 96.78<br>96.78<br>0.084 | 56.25<br>68.16<br>0.835 | **91.02**<br>**93.55**<br>**0.031** | **100.00**<br>**100.00**<br>**0.006** | 50.00<br>48.63<br>1.001 | 58.20<br>55.86<br>2.435 | 49.80<br>50.78<br>0.997 | **100.00**<br>**100.00**<br>**0.022** | 72.27<br>79.10<br>**0.128** |
| | Original → Mixed (Joint) | 98.33<br>98.33<br>0.047 | 59.18<br>73.44<br>0.740 | 84.77<br>90.82<br>0.073 | — | 50.00<br>46.48<br>1.020 | **79.88**<br>**81.84**<br>**0.237** | 50.20<br>54.30<br>0.982 | **100.00**<br>**100.00**<br>0.023 | 52.54<br>61.52<br>0.708 |
| | Original → Mixed (Frozen Encoder) | 96.44<br>96.44<br>0.062 | **98.63**<br>**99.80**<br>**0.101** | 81.84<br>89.26<br>0.213 | — | 73.63<br>83.59<br>0.595 | 54.49<br>64.26<br>0.867 | 52.93<br>57.81<br>0.606 | **100.00**<br>**100.00**<br>0.048 | 91.21<br>98.83<br>0.191 |
| DINO-WM | Original | 57.11<br>57.11<br>0.919 | 38.48<br>9.38<br>1.127 | 50.78<br>61.33<br>0.977 | — | 50.39<br>47.66<br>1.015 | 49.80<br>49.61<br>1.014 | 50.00<br>50.59<br>1.000 | 50.00<br>64.33<br>0.985 | 50.78<br>56.45<br>0.996 |
| | Mixed | 59.41<br>59.41<br>0.961 | 97.66<br>99.61<br>0.394 | 78.32<br>88.09<br>0.673 | 16.65<br>19.45<br>1.000 | 84.18<br>92.19<br>0.648 | 56.45<br>65.23<br>1.042 | 60.16<br>77.34<br>0.892 | **100.00**<br>**100.00**<br>0.217 | **100.00**<br>**100.00**<br>0.142 |
| | Original → Mixed | 84.78<br>84.78<br>0.411 | — | — | — | — | 55.27<br>61.13<br>1.111 | 62.89<br>80.27<br>0.871 | — | — |

where $E = \sum_q L_q$ and $B = \sum_q B_q$ are the total prediction and reference errors, respectively, with $B > 0$. Lower is better: $E/B = 0$ means exact predictions, while $E/B = 1$ matches the mean-future reference error. For equally weighted conditions sharing the current observation and future actions, this reference minimizes squared error among predictions that ignore the dynamics condition [18]; thus a deterministic history-independent model cannot achieve $E/B < 1$.

For the history diagnostic, we also report normalized history gain $G_H$. Let $\hat{z}_{qhct}$ be the prediction using history from condition $h$, while $z_{qkct}$ is the true future under condition $k$, with the same current observation and future actions. The aggregate matching- and mismatched-history errors are

$$E_+ = \sum_q \frac{1}{K_q C_q T} \sum_{k,c,t} \|\hat{z}_{qkct} - z_{qkct}\|^2,$$
$$E_- = \sum_q \frac{1}{K_q C_q T} \sum_{k,c,t} \frac{1}{K_q - 1} \sum_{h \neq k} \|\hat{z}_{qhct} - z_{qkct}\|^2. \tag{9}$$

The mismatched error averages equally over all $K_q - 1$ alternative histories. We compute

$$G_H = \frac{E_- - E_+}{B}, \tag{10}$$

using the same reference error $B$ as above. A positive $G_H$ means matching history reduces prediction error; unlike History win rate, it measures the size of this reduction and does not imply accurate prediction.

**Evaluation Scope and Reporting.** Adaptation Accuracy, History win rate, and NRE use Test for the main benchmark, as do multi-step prediction and additional mechanism analyses; training-query analyses use Training. The main task queries and extended multi-step evaluations differ in future actions and prediction horizons. Latent errors are measured in each checkpoint's own target representation and do not provide a common physical error scale across models. We report means and standard deviations across available training runs. For multi-step prediction, source-cluster bootstrap intervals reflect sampling uncertainty for the evaluated

checkpoints, not training-run variability (Appendix E). We separately report CEM planning success in the original environments to assess retention of existing control capabilities (Section 4).

# 4. Experiments

## 4.1. Experimental Setup

**Evaluated models and representation strategies.** We evaluate three action-conditioned latent world models with different representation-learning strategies: (1) DINO-WM [37], which trains a dynamics predictor on frozen, pretrained DINOv2 features; (2) PLDM [30], which jointly optimizes the visual encoder and dynamics predictor; and (3) LeWM [24], which combines latent prediction with representation regularization to mitigate representation collapse. All model weights remain fixed during evaluation. Because these models also differ in architecture and training objectives, their performance differences cannot be attributed to representation-learning strategies alone.

**Training schemes and design rationale.** We evaluate four training schemes to examine how training choices affect in-context dynamics adaptation (Table 2).
*Pretraining and data composition:* The Original scheme uses only fixed-dynamics data to assess adaptation without exposure to ContextWorld training data. The Mixed dataset combines original-environment and ContextWorld training windows at a $1:1$ sampling ratio. We compare training on Mixed from initialization with training on Mixed after an Original training stage. The latter includes an additional 10 epochs of Original training, so the two schemes do not have equal total training budgets.
*Visual encoder updates:* For LeWM and PLDM, we compare two fine-tuning schemes starting from the same Original checkpoint. Joint updates the visual encoder and the remaining trainable modules, whereas Frozen Encoder fixes the visual encoder while training the remaining modules. This comparison examines the effect of freezing the visual encoder on adaptation, but does not freeze the entire latent representation, as the projector remains trainable.

Models are trained independently for each task. We evaluate planning success in the original environments using the Cross-Entropy Method (CEM) [12] to assess whether existing control performance is maintained. Optimization details are provided in Appendix G.

## 4.2. Adaptation Results

Original models trained under fixed dynamics have future-selection accuracy near 50% on most of the seven binary tasks (Table 3). Adding ContextWorld training data improves several tasks, including arm mass for LeWM and PLDM, but the gains are not uniform.

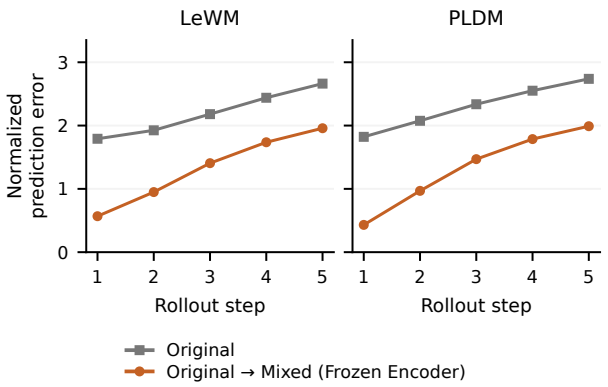



Figure 2. **Multi-step prediction on pusher displacement (Test).** Each rollout step spans five simulator steps without new observations. Prediction error is divided by the error of predicting the mean future representation across dynamics conditions. Ratios are computed within each checkpoint's representation, then averaged over runs. These extended queries differ from the one-step tasks; additional results appear in Appendix H.

Under Mixed training, LeWM and PLDM reach 97.70% and 100.00% on action delay, whereas DINO-WM reaches 16.65%. On pusher displacement, DINO-WM reaches 97.66%, outperforming both LeWM and PLDM under Mixed. All three models score at least 99.83% on door traversability. No model leads across all tasks.

On pusher displacement, Original $\rightarrow$ Mixed (Frozen Encoder) achieves Test History win rates of 98.83% for LeWM and 99.80% for PLDM, with NRE of 0.128 and 0.101. Yet their extended multi-step Test evaluations yield aggregate $E/B$ of 1.436 and 1.448, respectively (Figure 2). Thus, strong task-query adaptation does not guarantee low complete prediction error under different actions and longer horizons; these evaluations do not isolate the cause of the errors.

## 4.3. Effects of Training Choices

Freezing the pretrained visual encoder helps some tasks and hurts others. For LeWM, it improves contact friction Test accuracy from 50.39% to 86.13% relative to joint training, but reduces arm mass accuracy from 87.30% to 77.15% (Figure 3b; Table 3). Each comparison holds the data, initialization, and training duration fixed. Because projectors and action encoders remain trainable, this intervention does not isolate a static target representation.

Figure 3a examines the effect of data scaling on Test accuracy. For LeWM on pusher displacement, data scaling substantially benefits Original $\rightarrow$ Mixed (Joint), whereas Mixed stays near chance. PLDM improves under both schemes, with smaller accuracy gains than LeWM joint fine-tuning. NRE also improves with joint fine-tuning (Ap-

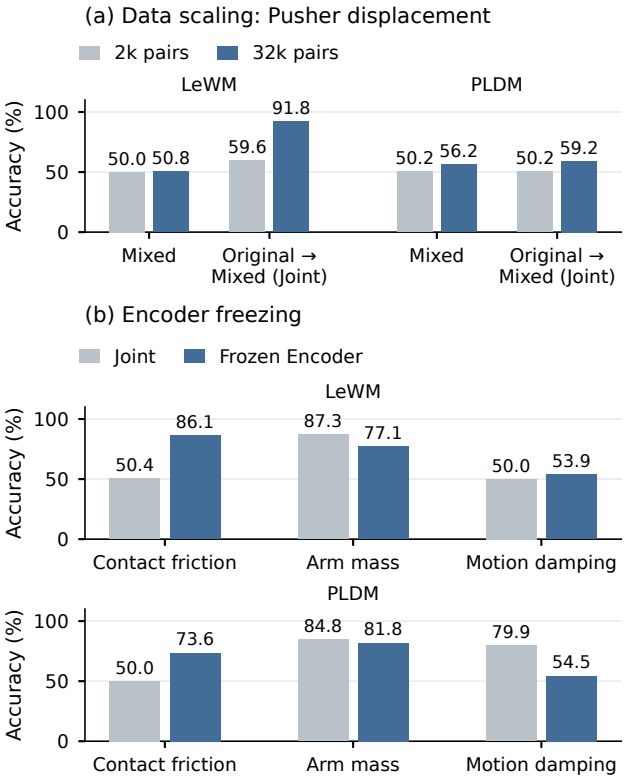



Figure 3. **Data scaling and encoder freezing on Test.** (a) LeWM and PLDM on pusher displacement with 2,048 versus 32,768 ContextWorld training pairs; original-environment data are unchanged. (b) Joint versus frozen-encoder fine-tuning on the same three tasks for both models. Each comparison uses the same Original checkpoint and task data. Bars show accuracy (%). Training epochs are fixed. Full results and run counts appear in Table 6 and Appendix H.1.

pendix H.1). Both data scales use the same number of training updates.

Under the tested 10-epoch recipe, increasing LeWM and PLDM from 18M to 183M parameters yields no consistent Test accuracy gain on friction and damping (Appendix H.2).

### 4.4. Prediction Loss and Adaptation

Weak evaluation performance may reflect poor generalization or conditional responses that were not learned well during training. We compare prediction loss with NRE on fixed training queries at epochs 5 and 10 (Figure 4).

On friction, PLDM joint training reduces prediction loss by 18.05%, yet training-query NRE remains near the zero-response reference. On damping with the encoder frozen, loss approaches convergence while NRE improves but remains high (Figure 5). Changing the scheme helps: freezing yields a stronger conditional response on friction, whereas joint training performs better on damping, with Test showing the same task-dependent ordering.

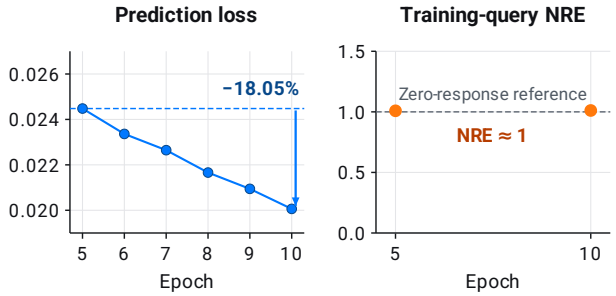



Figure 4. **Lower prediction loss does not ensure an accurate conditional response.** PLDM on contact friction, Original → Mixed (Joint). Recorded prediction loss falls by 18.05% from epochs 5 to 10, while training-query NRE stays near the zero-response reference of 1. Loss uses mixed training batches; NRE is measured on fixed paired training queries at epochs 5 and 10. Their adaptation accuracy is 49.51% and 49.71%, respectively. Full scheme and task comparisons appear in Figure 5.

These results expose a disconnect between prediction-loss optimization and history-conditioned adaptation, even on training data. Prediction loss aggregates errors across mixed data, whereas NRE directly measures error in the predicted differences between conditions. Optimizing the former does not ensure an accurate conditional response. This finding identifies insufficient constraints on history-conditioned dynamics in the evaluated training objectives and strategies.

## 5. Conclusion

ContextWorld establishes a standardized benchmark to evaluate in-context dynamics adaptation in action-conditioned latent visual world models under strictly frozen weights. Across nine tasks spanning four simulated environments, our paired history comparisons decouple whether models condition their predictions on interaction history from how accurately they forecast the resulting futures. Systematic evaluations of LeWM, PLDM, and DINO-WM demonstrate that while training on multi-dynamics data can improve adaptation, relative model advantages and the impact of visual encoder freezing remain task-dependent. Crucially, our training-distribution diagnostics expose a disconnect between prediction-loss optimization and in-context adaptation: conditional responses can remain poorly captured even as training loss steadily decreases or approaches convergence. These findings identify insufficient constraints on history-conditioned dynamics within the evaluated training objectives and strategies, underscoring the need for direct adaptation benchmarks and formulations that explicitly constrain history-conditioned responses.

Our findings are framed by specific diagnostic bound-

aries. Predictions are evaluated in each model's native representation rather than a shared physical error space; most tasks examine adaptation among dynamics conditions represented in training, while Speed evaluates unseen parameter values; and mixed-data checkpoints are trained independently per task, with planning experiments quantifying capability retention in unperturbed environments. These boundaries outline clear avenues for future research: establishing complementary cross-model evaluation in grounded physical coordinates, scaling toward generalist multi-task world models that generalize across broader dynamics, and examining closed-loop planning to test whether history-conditioned latent predictions translate directly into improved decision-making.

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## A. Input Sampling

Observations are typically sampled every five simulator steps. The actions executed between consecutive observations are included in the interaction history.

## B. Trajectory Construction and Pairing

**Continuous trajectories.** Each sample follows a continuous simulator trajectory from its history through its future, without intermediate resets or state overrides. Identical current observations may still conceal differences in velocity or contact state, so we perform task-specific checks of physical-state matching at the start of prediction. For action delay, we retain the actual queue of pending actions.

**Starting positions.** For tasks in which the two dynamics conditions cannot naturally share a starting position, we exchange their starting positions during generation and verify that the two conditions have identical sets of starting images. This check reduces the possibility that the starting image alone reveals the condition.

**Predictions without history.** Consider a complete, balanced pair with distinguishable encoded targets. A deterministic model that ignores history receives the same current observation and future action sequence for both samples and therefore produces the same prediction. This prediction cannot be strictly closer to each sample's own target than to the other target. Consequently, such a model scores at most 50% on the paired future-selection metric in Eq. 3; ties count as failures.

## C. Action Delay Response Groups

The delay data retain all 11 delays from 0 to 10 simulator steps. At the five-step horizon used for the main one-step score, delays of 5–10 produce the same stationary response and are grouped together. This score therefore uses six distinguishable response groups: 0, 1, 2, 3, 4, and 5–10.

## D. Dataset Statistics and Splits

### D.1. Dataset Sizes

Table 4 lists the task data used in the training and evaluation comparisons. Development supports method and checkpoint selection; Test is reserved for final evaluation. Training sizes exclude the original-environment data used for mixed training. A pair contains two trajectories with different dynamics conditions; a trajectory count refers to recorded episodes, not to the number of training windows sampled from them. An evaluation case fixes the current observation and future action sequence and can include histories and targets from several dynamics conditions.

The action delay counts refer to the dataset covering all delays from 0 to 10 simulator steps. Its Training split contains 5,120 base cases, each simulated under all 11 delays, giving 56,320 trajectories. Development and Test each contain 300 base cases, again evaluated under all 11 delays. Door Training contains 7,680 trajectories for each of the passable and blocked rules.

Several examples can share a source episode: pusher displacement can use multiple action windows from one episode, and forward and mirrored damping examples can share a source. The 10,000 gripper carry training pairs use 10,000 distinct source episodes.

### D.2. Paired Dynamics Conditions

Pusher displacement uses action gains of 60 and 140. Contact friction uses coefficients 0.05 and 0.80 between the pusher and the object. Motion damping fixes effective contact friction at 0.25 and uses simulator damping settings 0.2 and 1.0; here 0.2 gives faster decay, while 1.0 applies no additional decay. Arm mass uses densities of 500 and 1,500 for the MuJoCo arm link and finger geometry; the hand link retains its original density. Gripper carry changes whether the closed gripper carries the cube. Portal exit uses two exit rules for the same entrance, with paired trajectories reaching the shared query state before entering it. The numeric settings are simulator parameters, not additional model inputs.

### D.3. Speed Values and Door Positions

Table 5 lists the speed values used for Training and evaluation. Development and Test use the same evaluation speeds but different interaction trajectories. Each split contains 300 cases for each of the 14 evaluated speeds, or 4,200 cases in total before comparing histories from different speeds. The main task results use the unseen speeds within the training range.

Door position is specified by an integer coordinate along the wall. Test uses the 42 positions $30 + 4k$, where $k = 0, \ldots, 41$. Development uses the 16 positions 65, 77, 79, 87, 112, 115, 131, 152, 167, 169, 171, 172, 175, 180, 183, 185. Training uses the integers from 36 through 188 after excluding all Development and Test positions and the additional unused positions 40, 127, and 132, leaving 96 positions. The three position sets are disjoint. Both passable and blocked rules are used across the splits; the test distinction is the door position, not a new rule.

### D.4. Split Checks and Model Inputs

Split checks use source episode identifiers, trajectory template identifiers, or observation hashes, as appropriate for each task. Speed Development uses independently generated initial configurations and excludes initial states and current observations that duplicate Test cases. For Door, we

Table 4. Dataset sizes and condition coverage.

| Task | Training | Development | Test |
|---|---|---|---|
| Speed | 16,384 trajectories; 32 speeds | 300 cases per speed | 300 cases per speed |
| Pusher displacement | 32,768 pairs | 256 pairs | 256 pairs |
| Arm mass | 32,768 pairs | 256 pairs | 256 pairs |
| Action delay | 56,320 trajectories; 11 delays | 300 cases $\times$ 11 delays | 300 cases $\times$ 11 delays |
| Contact friction | 32,768 pairs | 256 pairs | 256 pairs |
| Motion damping | 32,768 pairs | 256 pairs | 256 pairs |
| Gripper carry rule | 10,000 pairs | 256 pairs | 256 pairs |
| Door traversability | 15,360 trajectories; 96 door positions | 300 cases; 16 door positions | 300 cases; 42 door positions |
| Portal exit | 32,768 pairs | 256 pairs | 256 pairs |

Table 5. Speed values in the benchmark. The four evaluation groups are used in both Development and Test. Interpolation values lie within the Training range but are absent from Training.

| Setting | Speed values |
|---|---|
| Training | 2.6, 2.7, 2.8, 2.9, 3.1, 3.3, 3.5, 3.7, 3.8, 3.9, 4.1, 4.2, 4.4, 4.5, 4.7, 5.0, 5.1, 5.3, 5.5, 5.7, 5.9, 6.1, 6.2, 6.3, 6.6, 6.7, 6.8, 7.0, 7.2, 7.3, 7.8, 7.9 |
| Seen speeds | 3.1, 5.1, 7.0 |
| Interpolation | 3.4, 4.8, 6.9 |
| Lower extrapolation | 1.75, 1.95, 2.15, 2.35 |
| Upper extrapolation | 8.25, 8.75, 9.5, 10.25 |

also check the position sets listed above for overlap.

Hidden dynamics values, simulator states, and pair identifiers are retained for data generation, validation, and scoring. They are not passed to the prediction model, which receives only the visual observations and actions defined in Sec. 3.1.

## E. Evaluation Details

**Paired tasks.** The strict comparisons in Eq. 3 do not count ties as correct. The two comparisons within a pair are dependent. Appendix B explains the upper bound for a deterministic model that ignores history.

**Speed.** For each target speed, accuracy is the proportion of cases in which matching history gives strictly lower prediction error than every alternative history. We then average accuracy equally across target speeds. Ties do not count as correct: if all histories produce the same prediction, accuracy is zero. The main results use the interpolation speeds listed in Appendix D.

**Action delay.** We select the candidate true future with the lowest mean squared error, resolving ties in favor of the smaller delay. The prediction is correct when the selected delay and the true delay belong to the same response group. Within each case, we first average correctness over delays in each group, then average equally across the six groups defined in Appendix C. The final accuracy is the mean over cases.

**Door traversability.** For a prediction made with matching history, we compare the mean squared errors to the passage and blockage futures. The prediction is correct only if the error to the true future is strictly smaller than the error to the alternative future. Accuracy is averaged over cases and the two rules; ties do not count as correct.

**Multi-step prediction horizons.** Each prediction step advances five simulator steps. We measure prediction errors at 5, 10, 15, 20, and 25 simulator steps after the current observation.

**Within-scene aggregation.** Let $z_{qkct}$ and $\hat{z}_{qkct}$ denote the true and predicted future representations for scene $q$, dynamics condition $k$, action sequence $c$, and prediction time $t$. With $K_q$ conditions, $C_q$ shared action sequences, and $T = 5$ prediction times, the errors used in Eq. 8 are

$$L_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|\hat{z}_{qkct} - z_{qkct}\|^2,$$

$$B_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|z_{qkct} - \bar{z}_{qct}\|^2,$$

$$\bar{z}_{qct} = \frac{1}{K_q} \sum_{k} z_{qkct}.$$

L2World tests history dependence by shuffling context observations [33]; our controls instead swap histories between dynamics conditions. For history controls, $E_+$ and $E_-$ sum the matching- and mismatched-history scene errors using the same averaging over conditions, action sequences, and

prediction times. When a sample has multiple mismatched histories, their errors are averaged equally. A positive $G_H$ indicates that matching history reduces error; it does not imply a low complete prediction error.

**Mean-future reference.** The reference in Eq. 8 is used only for scoring and is never provided to the model. The ratio is defined only when $\sum_q B_q > 0$. When conditions share the current observation and future actions and are equally weighted, a deterministic model that ignores history cannot achieve $E/B < 1$: it must make the same prediction across conditions, and the mean-future reference minimizes the error of such a prediction.

**Response error settings.** We specify the prediction horizon for each NRE analysis. The ratio in Eq. 6 is defined only when $\sum_i \|\Delta z_i\|^2 > 0$. NRE equal to 1 does not by itself show that a model ignores history. Errors shared by both predictions cancel in the difference, so an exact conditional response does not imply exact complete predictions.

**Uncertainty estimates.** For multi-step errors, we use cluster bootstrap resampling [15, 17] at the level of source trajectories or scenes, retaining related samples together. These intervals describe scene-sampling uncertainty for the evaluated checkpoints, not variability across training runs.

**Interpreting multiple metrics.** The metrics describe different aspects of the same predictions and are not independent pieces of evidence.

## F. Scoring Controls

We validate the paired scorer using constructed predictions on 256 Test pusher-displacement scenes, separately in the target representations of LeWM Original and Original → Mixed (Frozen Encoder). The queries use a nonzero action candidate and the five-step prediction horizon; the predictor is not run for this check. Both representations give the results below. With $m = (z_A + z_B)/2$, even predictions $m + 0.001(z_k - m)$ select the correct future while recovering only a tiny fraction of the true response. This control explains why we report response error alongside task accuracy. It verifies the scorer on these queries, not physical calibration across all representations.

| Constructed prediction | Accuracy (%) | History win rate (%) | NRE |
|---|---|---|---|
| True target for each condition | 100 | 100 | 0 |
| Same target $z_A$ for both conditions | 50 | 0 | 1 |
| Same midpoint $m$ for both conditions | 0 | 0 | 1 |
| Swapped true targets | 0 | 0 | 4 |
| $m + 0.001(z_k - m)$ for condition $k$ | 100 | 100 | 0.998001 |

## G. Training and Diagnostic Settings

**Optimization.** The training recipe specifies 10 epochs per stage on eight GPUs, with 128 windows per device and no gradient accumulation, giving a global batch size of 1,024. LeWM and PLDM use AdamW with learning rate $5 \times 10^{-5}$ and weight decay 0.001. DINO-WM uses learning rate $5 \times 10^{-4}$ and zero weight decay; its pretrained DINOv2-small visual encoder remains frozen. The models retain their respective prediction and regularization losses. The Original → Mixed schemes load model weights but reset optimizer state. Encoder freezing fixes both parameters and training-state buffers; it does not freeze the projector or action encoder.

**Input and sampling.** The standard setup uses three observation frames, five simulator steps between frames, and a one-frame prediction target. Action delay mixed training instead uses seven-frame history. The original-environment data are the TwoRoom, PushT, Reacher, and Cube datasets released with LeWM; each task uses only the dataset for its environment. Only the ContextWorld Training split is mixed into training. For component lengths $N_{\mathrm{orig}}$ and $N_{\mathrm{CW}}$, the mixture has a nominal epoch length of $2\max(N_{\mathrm{orig}}, N_{\mathrm{CW}})$ windows before partitioning and batching, with equal counts from the two components. The smaller component is sampled repeatedly. Consequently, equal epoch counts do not guarantee equal update counts across data scales, and the two-stage schemes include additional original-environment training.

**Prediction diagnostics.** The main task comparison uses Test. The multi-step and history diagnostics below use Test and the same checkpoints as the main task comparison. Multi-step evaluation covers 2,436 source scenes across all nine tasks and predicts at 5, 10, 15, 20, and 25 simulator steps using only the initial observed history. The candidate actions and multi-step targets extend the one-step task queries. For gripper carry, we replay all 256 held-out source cases and render their histories and extended futures together, preserving the paired current-observation constraint without mixing stored images with newly rendered futures. These coherent replays retain the source conditions and actions but do not reproduce the stored images exactly; the original task-query evaluation is unchanged. Figure 2 reports $E_t/B_t$: prediction and mean-future errors are summed over queries separately at each horizon, normalized within each checkpoint, then averaged equally over training repetitions. The aggregate $E/B$ instead combines errors over all horizons before division; it is not the arithmetic mean of these per-horizon ratios. Each checkpoint is scored separately, and matching training repetitions are then averaged equally. Intervals use 4,000 source-cluster bootstrap samples; all conditions, actions, and times from a source remain together. They condition on the evaluated checkpoints and do not estimate variation from retraining the model. Original action-delay checkpoints do not have the native seven-frame interface and are excluded from this comparison. DINO-WM has no jointly updated visual-encoder scheme; its comparable Original → Mixed results cover Speed, damping, and gripper carry.

**Planning evaluation.** CEM results measure success in the original environments, with no hidden-dynamics history intervention. Most checkpoint evaluations use six evaluation seeds with 50 episodes each; some original LeWM and PLDM references use three sets of 100 episodes or an aggregated 300-episode result. Evaluation repetitions are pooled within each checkpoint, then training repetitions are averaged equally. A single original-environment checkpoint supplies the reference for all tasks from that environment; these repeated entries are not independent experiments.

**Training-query analysis.** The friction and damping comparisons use 32,768 training pairs, identical original-environment initialization within each task, and the same 50/50 mixture, batch size, and duration. We evaluate 512 fixed training pairs and 256 Test pairs per task at epochs 5 and 10, predicting the next observation five simulator steps ahead. Loss summaries use the logged mixed-training batches, approximately 50–70 records per epoch, rather than every training batch. Checkpoints and logs are aligned by update step. The curves show recorded means and batch interquartile ranges; loss medians give the same monotonic direction for friction joint training from epochs 5 to 10. NRE is evaluated in each checkpoint's target representation and is not a common physical error across checkpoints. For damping with the encoder frozen, prediction loss changes by only $-0.39\%$ between these endpoints and fluctuates in between. The late-training plateau is consistent with the loss approaching convergence.

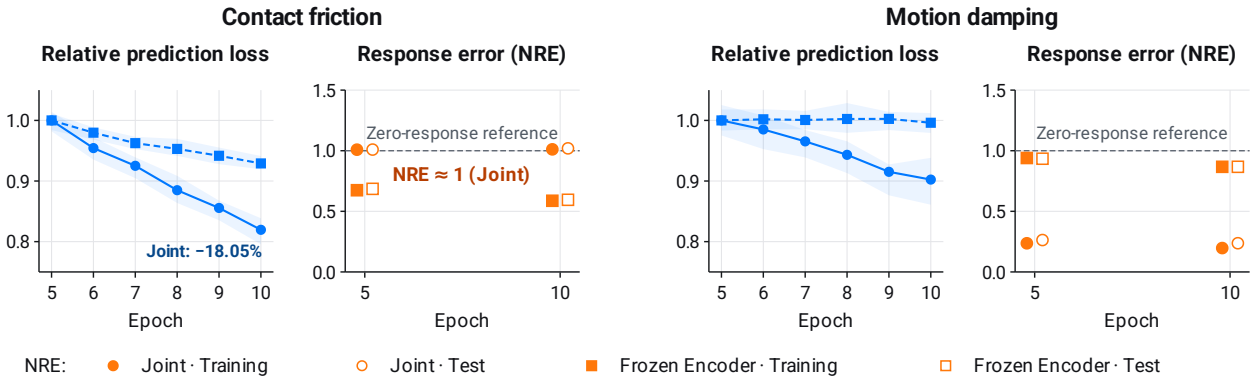



Figure 5. **Training loss and conditional responses across tasks and schemes.** Within each task, PLDM Joint and Frozen Encoder share Original initialization. Loss is normalized to epoch 5; shading shows batch interquartile ranges. Blue denotes prediction loss and orange denotes NRE; circles denote Joint and squares denote Frozen Encoder. NRE uses filled markers for Training and open markers for Test, evaluated only at epochs 5 and 10. Small horizontal offsets separate the two splits; 1 marks the zero-response reference. Loss uses mixed batches, whereas NRE uses fixed paired task queries.

## H. Additional Results

Table 6 reports all three Test adaptation metrics with training-run counts and standard deviations. Table 7 combines Test history and response metrics with multi-step prediction errors and original-environment planning results. History win rate measures strict error reduction from matching history on the task queries; NRE measures the conditional response on those queries. $E/B$ uses the extended multi-step action panels and includes errors shared by both conditions. Because the query sets and prediction horizons differ, these columns should not be treated as measurements of the same event. All latent errors are computed in the checkpoint's own target representation.

Table 6. **Complete Test adaptation metrics.** Each row uses the same evaluated checkpoints for all three metrics. Entries are means $\pm$ sample standard deviations across $n$ training runs; a single run has no sample standard deviation. Accuracy and History win rate are percentages; NRE is dimensionless. Speed uses unseen interpolation speeds, and its Accuracy and History win rate use the same scoring rule. Action delay uses seven-frame history. Only available comparable configurations are listed.

| Task | Model | Training scheme | $n$ | Accuracy ↑ | History win rate ↑ | NRE ↓ |
|---|---|---|---|---|---|---|
| Speed | LeWM | Original | 3 | $58.04 \pm 1.70$ | $58.04 \pm 1.70$ | $0.896 \pm 0.027$ |
| Speed | LeWM | Mixed | 3 | $98.19 \pm 0.57$ | $98.19 \pm 0.57$ | $0.042 \pm 0.011$ |
| Speed | LeWM | Original → Mixed (Joint) | 1 | 96.89 | 96.89 | 0.055 |
| Speed | LeWM | Original → Mixed (Frozen Encoder) | 1 | 98.44 | 98.44 | 0.030 |
| Speed | PLDM | Original | 3 | $48.41 \pm 4.33$ | $48.41 \pm 4.33$ | $0.922 \pm 0.026$ |
| Speed | PLDM | Mixed | 3 | $96.78 \pm 2.02$ | $96.78 \pm 2.02$ | $0.084 \pm 0.013$ |
| Speed | PLDM | Original → Mixed (Joint) | 1 | 98.33 | 98.33 | 0.047 |

| Task | Model | Training scheme | $n$ | Accuracy ↑ | History win rate ↑ | NRE ↓ |
|---|---|---|---|---|---|---|
| Speed | PLDM | Original → Mixed (Frozen Encoder) | 1 | 96.44 | 96.44 | 0.062 |
| Speed | DINO-WM | Original | 1 | 57.11 | 57.11 | 0.919 |
| Speed | DINO-WM | Mixed | 3 | $59.41 \pm 3.87$ | $59.41 \pm 3.87$ | $0.961 \pm 0.011$ |
| Speed | DINO-WM | Original → Mixed | 1 | 84.78 | 84.78 | 0.411 |
| Pusher displacement | LeWM | Original | 3 | $48.96 \pm 1.00$ | $28.32 \pm 12.76$ | $1.018 \pm 0.016$ |
| Pusher displacement | LeWM | Mixed | 1 | 50.78 | 79.10 | 0.974 |
| Pusher displacement | LeWM | Original → Mixed (Joint) | 1 | 91.80 | 99.02 | 0.215 |
| Pusher displacement | LeWM | Original → Mixed (Frozen Encoder) | 1 | 97.46 | 98.83 | 0.128 |
| Pusher displacement | PLDM | Original | 3 | $45.57 \pm 1.88$ | $9.38 \pm 4.20$ | $1.058 \pm 0.012$ |
| Pusher displacement | PLDM | Mixed | 1 | 56.25 | 68.16 | 0.835 |
| Pusher displacement | PLDM | Original → Mixed (Joint) | 1 | 59.18 | 73.44 | 0.740 |
| Pusher displacement | PLDM | Original → Mixed (Frozen Encoder) | 1 | 98.63 | 99.80 | 0.101 |
| Pusher displacement | DINO-WM | Original | 1 | 38.48 | 9.38 | 1.127 |
| Pusher displacement | DINO-WM | Mixed | 1 | 97.66 | 99.61 | 0.394 |
| Arm mass | LeWM | Original | 3 | $50.07 \pm 0.11$ | $51.63 \pm 0.60$ | $0.977 \pm 0.002$ |
| Arm mass | LeWM | Mixed | 1 | 85.74 | 89.84 | 0.045 |
| Arm mass | LeWM | Original → Mixed (Joint) | 1 | 87.30 | 93.36 | 0.038 |
| Arm mass | LeWM | Original → Mixed (Frozen Encoder) | 1 | 77.15 | 86.52 | 0.346 |
| Arm mass | PLDM | Original | 3 | $51.82 \pm 1.30$ | $66.99 \pm 9.68$ | $0.921 \pm 0.016$ |
| Arm mass | PLDM | Mixed | 1 | 91.02 | 93.55 | 0.031 |
| Arm mass | PLDM | Original → Mixed (Joint) | 1 | 84.77 | 90.82 | 0.073 |
| Arm mass | PLDM | Original → Mixed (Frozen Encoder) | 1 | 81.84 | 89.26 | 0.213 |
| Arm mass | DINO-WM | Original | 1 | 50.78 | 61.33 | 0.977 |
| Arm mass | DINO-WM | Mixed | 1 | 78.32 | 88.09 | 0.673 |
| Action delay | LeWM | Mixed | 3 | $97.70 \pm 0.59$ | $99.18 \pm 0.38$ | $0.013 \pm 0.002$ |
| Action delay | PLDM | Mixed | 3 | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $0.006 \pm 0.001$ |
| Action delay | DINO-WM | Mixed | 3 | $16.65 \pm 0.03$ | $19.45 \pm 0.11$ | $1.000 \pm 0.000$ |
| Contact friction | LeWM | Original | 3 | $49.93 \pm 0.23$ | $50.13 \pm 0.63$ | $1.001 \pm 0.004$ |
| Contact friction | LeWM | Mixed | 1 | 50.00 | 49.41 | 1.000 |
| Contact friction | LeWM | Original → Mixed (Joint) | 1 | 50.39 | 54.30 | 0.999 |
| Contact friction | LeWM | Original → Mixed (Frozen Encoder) | 1 | 86.13 | 92.77 | 0.429 |
| Contact friction | PLDM | Original | 3 | $50.00 \pm 0.34$ | $49.09 \pm 1.58$ | $1.014 \pm 0.010$ |
| Contact friction | PLDM | Mixed | 1 | 50.00 | 48.63 | 1.001 |
| Contact friction | PLDM | Original → Mixed (Joint) | 1 | 50.00 | 46.48 | 1.020 |
| Contact friction | PLDM | Original → Mixed (Frozen Encoder) | 1 | 73.63 | 83.59 | 0.595 |
| Contact friction | DINO-WM | Original | 1 | 50.39 | 47.66 | 1.015 |
| Contact friction | DINO-WM | Mixed | 1 | 84.18 | 92.19 | 0.648 |

| Task | Model | Training scheme | $n$ | Accuracy ↑ | History win rate ↑ | NRE ↓ |
|---|---|---|---|---|---|---|
| Motion damping | LeWM | Original | 3 | $50.00 \pm 0.00$ | $48.63 \pm 1.60$ | $1.007 \pm 0.009$ |
| Motion damping | LeWM | Mixed | 1 | 50.00 | 48.24 | 1.004 |
| Motion damping | LeWM | Original → Mixed (Joint) | 1 | 50.00 | 49.61 | 1.001 |
| Motion damping | LeWM | Original → Mixed (Frozen Encoder) | 1 | 53.91 | 61.91 | 0.958 |
| Motion damping | PLDM | Original | 3 | $50.26 \pm 0.30$ | $50.52 \pm 0.63$ | $0.947 \pm 0.007$ |
| Motion damping | PLDM | Mixed | 1 | 58.20 | 55.86 | 2.435 |
| Motion damping | PLDM | Original → Mixed (Joint) | 1 | 79.88 | 81.84 | 0.237 |
| Motion damping | PLDM | Original → Mixed (Frozen Encoder) | 1 | 54.49 | 64.26 | 0.867 |
| Motion damping | DINO-WM | Original | 1 | 49.80 | 49.61 | 1.014 |
| Motion damping | DINO-WM | Mixed | 1 | 56.45 | 65.23 | 1.042 |
| Motion damping | DINO-WM | Original → Mixed | 1 | 55.27 | 61.13 | 1.111 |
| Gripper carry rule | LeWM | Original | 3 | $52.21 \pm 1.11$ | $73.76 \pm 1.97$ | $0.909 \pm 0.029$ |
| Gripper carry rule | LeWM | Mixed | 1 | 50.00 | 49.80 | 1.000 |
| Gripper carry rule | LeWM | Original → Mixed (Joint) | 1 | 50.00 | 52.34 | 0.997 |
| Gripper carry rule | LeWM | Original → Mixed (Frozen Encoder) | 1 | 76.37 | 84.57 | 0.271 |
| Gripper carry rule | PLDM | Original | 3 | $50.78 \pm 0.20$ | $57.42 \pm 1.35$ | $0.937 \pm 0.007$ |
| Gripper carry rule | PLDM | Mixed | 1 | 49.80 | 50.78 | 0.997 |
| Gripper carry rule | PLDM | Original → Mixed (Joint) | 1 | 50.20 | 54.30 | 0.982 |
| Gripper carry rule | PLDM | Original → Mixed (Frozen Encoder) | 1 | 52.93 | 57.81 | 0.606 |
| Gripper carry rule | DINO-WM | Original | 1 | 50.00 | 50.59 | 1.000 |
| Gripper carry rule | DINO-WM | Mixed | 1 | 60.16 | 77.34 | 0.892 |
| Gripper carry rule | DINO-WM | Original → Mixed | 1 | 62.89 | 80.27 | 0.871 |
| Door traversability | LeWM | Original | 3 | $50.33 \pm 0.00$ | $50.94 \pm 2.84$ | $0.996 \pm 0.003$ |
| Door traversability | LeWM | Mixed | 3 | $99.83 \pm 0.29$ | $100.00 \pm 0.00$ | $0.041 \pm 0.018$ |
| Door traversability | LeWM | Original → Mixed (Joint) | 1 | 99.83 | 100.00 | 0.038 |
| Door traversability | LeWM | Original → Mixed (Frozen Encoder) | 1 | 100.00 | 100.00 | 0.036 |
| Door traversability | PLDM | Original | 3 | $50.06 \pm 0.10$ | $51.50 \pm 0.67$ | $0.997 \pm 0.002$ |
| Door traversability | PLDM | Mixed | 3 | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $0.022 \pm 0.005$ |
| Door traversability | PLDM | Original → Mixed (Joint) | 1 | 100.00 | 100.00 | 0.023 |
| Door traversability | PLDM | Original → Mixed (Frozen Encoder) | 1 | 100.00 | 100.00 | 0.048 |
| Door traversability | DINO-WM | Original | 1 | 50.00 | 64.33 | 0.985 |
| Door traversability | DINO-WM | Mixed | 3 | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $0.217 \pm 0.003$ |
| Portal exit | LeWM | Original | 3 | $50.07 \pm 0.11$ | $51.95 \pm 0.70$ | $0.990 \pm 0.003$ |
| Portal exit | LeWM | Mixed | 1 | 50.78 | 76.37 | 0.922 |
| Portal exit | LeWM | Original → Mixed (Joint) | 1 | 50.39 | 66.80 | 0.934 |
| Portal exit | LeWM | Original → Mixed (Frozen Encoder) | 1 | 77.93 | 94.34 | 0.264 |
| Portal exit | PLDM | Original | 3 | $50.07 \pm 0.11$ | $51.50 \pm 0.30$ | $0.996 \pm 0.001$ |
| Portal exit | PLDM | Mixed | 1 | 72.27 | 79.10 | 0.128 |
| Portal exit | PLDM | Original → Mixed (Joint) | 1 | 52.54 | 61.52 | 0.708 |
| Portal exit | PLDM | Original → Mixed (Frozen Encoder) | 1 | 91.21 | 98.83 | 0.191 |
| Portal exit | DINO-WM | Original | 1 | 50.78 | 56.45 | 0.996 |
| Portal exit | DINO-WM | Mixed | 1 | 100.00 | 100.00 | 0.142 |

Table 7. Test prediction diagnostics and original-environment CEM. Entries average the available training repetitions; $E/B$ intervals are 95% source-bootstrap intervals. History win rate and CEM are percentages. Scheme names match the main text.

| Task | Model | Scheme | History win rate ↑ | NRE ↓ | $E/B$ ↓ | $E/B$ interval | CEM ↑ |
|---|---|---|---|---|---|---|---|
| Speed | LeWM | Original | 58.04 | 0.896 | 1.512 | [1.491, 1.533] | 92.44 |
| Speed | LeWM | Mixed | 98.19 | 0.042 | 0.368 | [0.350, 0.388] | 97.44 |
| Speed | LeWM | Original → Mixed (Joint) | 96.89 | 0.055 | 0.399 | [0.376, 0.423] | 97.00 |
| Speed | LeWM | Original → Mixed (Frozen Encoder) | 98.44 | 0.030 | 0.145 | [0.133, 0.157] | 95.00 |
| Speed | PLDM | Original | 48.41 | 0.922 | 3.446 | [3.366, 3.542] | 90.56 |
| Speed | PLDM | Mixed | 96.78 | 0.084 | 0.515 | [0.492, 0.539] | 96.89 |
| Speed | PLDM | Original → Mixed (Joint) | 98.33 | 0.047 | 0.380 | [0.355, 0.408] | 96.33 |
| Speed | PLDM | Original → Mixed (Frozen Encoder) | 96.44 | 0.062 | 0.308 | [0.290, 0.326] | 98.00 |
| Speed | DINO-WM | Original | 57.11 | 0.919 | 1.967 | [1.949, 1.987] | 100.00 |
| Speed | DINO-WM | Mixed | 59.41 | 0.961 | 2.086 | [2.054, 2.123] | 99.89 |
| Speed | DINO-WM | Original → Mixed | 84.78 | 0.411 | 0.886 | [0.851, 0.924] | 100.00 |
| Pusher displacement | LeWM | Original | 28.32 | 1.018 | 2.279 | [2.211, 2.354] | 82.22 |
| Pusher displacement | LeWM | Mixed | 79.10 | 0.974 | 3.779 | [3.596, 3.968] | 78.67 |
| Pusher displacement | LeWM | Original → Mixed (Joint) | 99.02 | 0.215 | 2.426 | [2.353, 2.504] | 86.00 |
| Pusher displacement | LeWM | Original → Mixed (Frozen Encoder) | 98.83 | 0.128 | 1.436 | [1.365, 1.510] | 73.67 |
| Pusher displacement | PLDM | Original | 9.38 | 1.058 | 2.386 | [2.332, 2.442] | 75.67 |
| Pusher displacement | PLDM | Mixed | 68.16 | 0.835 | 6.284 | [6.001, 6.589] | 61.33 |
| Pusher displacement | PLDM | Original → Mixed (Joint) | 73.44 | 0.740 | 3.720 | [3.604, 3.840] | 72.00 |
| Pusher displacement | PLDM | Original → Mixed (Frozen Encoder) | 99.80 | 0.101 | 1.448 | [1.397, 1.501] | 59.33 |
| Pusher displacement | DINO-WM | Original | 9.38 | 1.127 | 3.140 | [3.024, 3.261] | 76.67 |
| Pusher displacement | DINO-WM | Mixed | 99.61 | 0.394 | 1.931 | [1.884, 1.981] | 76.67 |
| Arm mass | LeWM | Original | 51.63 | 0.977 | 741.737 | [644.066, 844.093] | 55.89 |
| Arm mass | LeWM | Mixed | 89.84 | 0.045 | 51.802 | [47.895, 56.217] | 58.00 |
| Arm mass | LeWM | Original → Mixed (Joint) | 93.36 | 0.038 | 47.590 | [44.059, 51.609] | 59.00 |
| Arm mass | LeWM | Original → Mixed (Frozen Encoder) | 86.52 | 0.346 | 33.070 | [30.472, 35.587] | 59.67 |
| Arm mass | PLDM | Original | 66.99 | 0.921 | 19.698 | [17.799, 21.952] | 69.67 |
| Arm mass | PLDM | Mixed | 93.55 | 0.031 | 13.815 | [12.861, 14.967] | 51.67 |
| Arm mass | PLDM | Original → Mixed (Joint) | 90.82 | 0.073 | 46.000 | [42.230, 50.596] | 58.67 |
| Arm mass | PLDM | Original → Mixed (Frozen Encoder) | 89.26 | 0.213 | 17.048 | [16.034, 18.176] | 89.00 |
| Arm mass | DINO-WM | Original | 61.33 | 0.977 | 49.483 | [46.626, 52.747] | 90.67 |
| Arm mass | DINO-WM | Mixed | 88.09 | 0.673 | 14.166 | [13.764, 14.537] | 75.00 |
| Action delay | LeWM | Mixed | 99.18 | 0.013 | 1.429 | [1.406, 1.453] | 87.33 |

| Task | Model | Scheme | History win rate ↑ | NRE ↓ | $E/B$ ↓ | $E/B$ interval | CEM ↑ |
|---|---|---|---|---|---|---|---|
| Action delay | PLDM | Mixed | 100.00 | 0.006 | 1.818 | [1.804, 1.832] | 90.89 |
| Action delay | DINO-WM | Mixed | 19.45 | 1.000 | 66.094 | [65.265, 66.948] | 33.56 |
| Contact friction | LeWM | Original | 50.13 | 1.001 | 2.716 | [2.642, 2.791] | 82.22 |
| Contact friction | LeWM | Mixed | 49.41 | 1.000 | 2.727 | [2.678, 2.781] | 76.67 |
| Contact friction | LeWM | Original → Mixed (Joint) | 54.30 | 0.999 | 2.717 | [2.677, 2.757] | 86.00 |
| Contact friction | LeWM | Original → Mixed (Frozen Encoder) | 92.77 | 0.429 | 2.902 | [2.843, 2.960] | 83.33 |
| Contact friction | PLDM | Original | 49.09 | 1.014 | 3.809 | [3.686, 3.940] | 75.67 |
| Contact friction | PLDM | Mixed | 48.63 | 1.001 | 6.392 | [6.077, 6.741] | 65.67 |
| Contact friction | PLDM | Original → Mixed (Joint) | 46.48 | 1.020 | 7.236 | [6.961, 7.541] | 73.33 |
| Contact friction | PLDM | Original → Mixed (Frozen Encoder) | 83.59 | 0.595 | 4.536 | [4.408, 4.666] | 79.67 |
| Contact friction | DINO-WM | Original | 47.66 | 1.015 | 6.252 | [5.971, 6.531] | 76.67 |
| Contact friction | DINO-WM | Mixed | 92.19 | 0.648 | 3.695 | [3.611, 3.776] | 75.67 |
| Motion damping | LeWM | Original | 48.63 | 1.007 | 5.341 | [5.105, 5.611] | 82.22 |
| Motion damping | LeWM | Mixed | 48.24 | 1.004 | 248.446 | [216.917, 288.116] | 78.00 |
| Motion damping | LeWM | Original → Mixed (Joint) | 49.61 | 1.001 | 87.139 | [77.847, 98.646] | 88.33 |
| Motion damping | LeWM | Original → Mixed (Frozen Encoder) | 61.91 | 0.958 | 8.316 | [7.878, 8.794] | 84.33 |
| Motion damping | PLDM | Original | 50.52 | 0.947 | 6.216 | [5.937, 6.535] | 75.67 |
| Motion damping | PLDM | Mixed | 55.86 | 2.435 | 21.427 | [18.508, 25.342] | 61.00 |
| Motion damping | PLDM | Original → Mixed (Joint) | 81.84 | 0.237 | 8.323 | [8.060, 8.597] | 70.00 |
| Motion damping | PLDM | Original → Mixed (Frozen Encoder) | 64.26 | 0.867 | 5.680 | [5.431, 5.959] | 78.33 |
| Motion damping | DINO-WM | Original | 49.61 | 1.014 | 8.442 | [8.184, 8.713] | 76.67 |
| Motion damping | DINO-WM | Mixed | 65.23 | 1.042 | 7.292 | [7.117, 7.484] | 76.67 |
| Motion damping | DINO-WM | Original → Mixed | 61.13 | 1.111 | 6.975 | [6.800, 7.165] | 76.00 |
| Gripper carry rule | LeWM | Original | 73.76 | 0.909 | 10.647 | [9.807, 11.634] | 65.44 |
| Gripper carry rule | LeWM | Mixed | 49.80 | 1.000 | 170.794 | [132.250, 222.804] | 61.33 |
| Gripper carry rule | LeWM | Original → Mixed (Joint) | 52.34 | 0.997 | 12.148 | [10.850, 13.816] | 71.00 |

| Task | Model | Scheme | History win rate ↑ | NRE ↓ | $E/B$ ↓ | $E/B$ interval | CEM ↑ |
|---|---|---|---|---|---|---|---|
| Gripper carry rule | LeWM | Original → Mixed (Frozen Encoder) | 84.57 | 0.271 | 11.149 | [10.363, 12.008] | 67.00 |
| Gripper carry rule | PLDM | Original | 57.42 | 0.937 | 11.994 | [9.809, 15.328] | 53.44 |
| Gripper carry rule | PLDM | Mixed | 50.78 | 0.997 | 138.687 | [115.564, 166.941] | 55.00 |
| Gripper carry rule | PLDM | Original → Mixed (Joint) | 54.30 | 0.982 | 26.192 | [23.629, 29.169] | 56.67 |
| Gripper carry rule | PLDM | Original → Mixed (Frozen Encoder) | 57.81 | 0.606 | 25.681 | [20.138, 33.793] | 56.67 |
| Gripper carry rule | DINO-WM | Original | 50.59 | 1.000 | 40.837 | [39.662, 42.051] | 72.67 |
| Gripper carry rule | DINO-WM | Mixed | 77.34 | 0.892 | 11.402 | [11.088, 11.709] | 71.67 |
| Gripper carry rule | DINO-WM | Original → Mixed | 80.27 | 0.871 | 11.286 | [10.982, 11.585] | 73.67 |
| Door traversability | LeWM | Original | 50.94 | 0.996 | 5.255 | [5.151, 5.364] | 92.44 |
| Door traversability | LeWM | Mixed | 100.00 | 0.041 | 3.027 | [2.984, 3.070] | 95.11 |
| Door traversability | LeWM | Original → Mixed (Joint) | 100.00 | 0.038 | 2.309 | [2.261, 2.358] | 97.67 |
| Door traversability | LeWM | Original → Mixed (Frozen Encoder) | 100.00 | 0.036 | 2.495 | [2.459, 2.528] | 94.00 |
| Door traversability | PLDM | Original | 51.50 | 0.997 | 8.278 | [8.069, 8.487] | 90.56 |
| Door traversability | PLDM | Mixed | 100.00 | 0.022 | 2.370 | [2.352, 2.389] | 95.78 |
| Door traversability | PLDM | Original → Mixed (Joint) | 100.00 | 0.023 | 2.352 | [2.318, 2.387] | 99.33 |
| Door traversability | PLDM | Original → Mixed (Frozen Encoder) | 100.00 | 0.048 | 2.449 | [2.417, 2.479] | 98.67 |
| Door traversability | DINO-WM | Original | 64.33 | 0.985 | 3.812 | [3.710, 3.915] | 100.00 |
| Door traversability | DINO-WM | Mixed | 100.00 | 0.217 | 1.261 | [1.238, 1.286] | 97.56 |
| Portal exit | LeWM | Original | 51.95 | 0.990 | 908.548 | [650.682, 1395.382] | 92.44 |
| Portal exit | LeWM | Mixed | 76.37 | 0.922 | 47.098 | [35.158, 68.065] | 98.33 |
| Portal exit | LeWM | Original → Mixed (Joint) | 66.80 | 0.934 | 49.651 | [34.092, 86.154] | 95.67 |
| Portal exit | LeWM | Original → Mixed (Frozen Encoder) | 94.34 | 0.264 | 11.124 | [9.727, 13.437] | 89.33 |
| Portal exit | PLDM | Original | 51.50 | 0.996 | 1067.079 | [840.093, 1434.396] | 90.56 |
| Portal exit | PLDM | Mixed | 79.10 | 0.128 | 36.403 | [33.836, 39.033] | 97.00 |
| Portal exit | PLDM | Original → Mixed (Joint) | 61.52 | 0.708 | 23.385 | [21.534, 25.531] | 98.67 |
| Portal exit | PLDM | Original → Mixed (Frozen Encoder) | 98.83 | 0.191 | 10.203 | [9.431, 11.154] | 98.67 |
| Portal exit | DINO-WM | Original | 56.45 | 0.996 | 42.346 | [41.223, 43.577] | 100.00 |
| Portal exit | DINO-WM | Mixed | 100.00 | 0.142 | 8.046 | [7.960, 8.133] | 100.00 |

## H.1. Data Scale

Figure 3a compares LeWM and PLDM on pusher displacement using the same Test trajectories at both training-data scales. Table 8 reports all three adaptation metrics for these checkpoints. All data-scale comparisons hold training epochs fixed. For pusher displacement, the original-data component determines the mixture length at both 2,048 and 32,768 pairs, giving 3,483 training batches per epoch. The two-stage scheme uses 17,410 Original-pretraining batches followed by 34,830 Mixed batches, compared with 34,830 batches for Mixed training from scratch.

Table 8. LeWM and PLDM data scaling on pusher displacement (Test). Entries are means over $n$ training runs; pair counts refer to ContextWorld training data.

| Model | Scheme | Pairs | $n$ | Accuracy (%) | History win rate (%) | NRE |
|---|---|---|---|---|---|---|
| LeWM | Mixed | 2,048 | 3 | 50.00 | 51.63 | 1.000 |
| LeWM | Mixed | 32,768 | 1 | 50.78 | 79.10 | 0.974 |
| LeWM | Original → Mixed (Joint) | 2,048 | 1 | 59.57 | 83.01 | 0.813 |
| LeWM | Original → Mixed (Joint) | 32,768 | 1 | 91.80 | 99.02 | 0.215 |
| PLDM | Mixed | 2,048 | 3 | 50.20 | 50.65 | 1.005 |
| PLDM | Mixed | 32,768 | 1 | 56.25 | 68.16 | 0.835 |
| PLDM | Original → Mixed (Joint) | 2,048 | 1 | 50.20 | 49.22 | 1.001 |
| PLDM | Original → Mixed (Joint) | 32,768 | 1 | 59.18 | 73.44 | 0.740 |

Table 9 reports the remaining data-scale comparisons on Test. The sample counts describe pairs, not necessarily independent source episodes. Pusher displacement uses 2,048, 8,192, and 32,768 pairs; friction and damping compare 8,192 with 32,768; mass and portal exit compare 2,048 with 32,768. Gripper carry expands from 2,048 to 10,000 source episodes. Portal expansion also broadens initial distances and action magnitudes, so its comparison changes coverage as well as size.

Table 9. Data-scale comparisons on Test. $n$ counts training repetitions contributing to task accuracy and NRE. Planning repetitions can differ and are not counted as additional training runs.

| Task | Model | Scheme | Pairs | $n$ | Accuracy (%) | NRE |
|---|---|---|---|---|---|---|
| Pusher displacement | LeWM | Mixed | 2,048 | 3 | 50.00 | 1.000 |
| Pusher displacement | LeWM | Mixed | 32,768 | 1 | 50.78 | 0.974 |
| Pusher displacement | LeWM | Original → Mixed (Joint) | 2,048 | 1 | 59.57 | 0.813 |
| Pusher displacement | LeWM | Original → Mixed (Joint) | 8,192 | 1 | 83.40 | 0.377 |
| Pusher displacement | LeWM | Original → Mixed (Joint) | 32,768 | 1 | 91.80 | 0.215 |
| Pusher displacement | PLDM | Mixed | 2,048 | 3 | 50.20 | 1.005 |
| Pusher displacement | PLDM | Mixed | 32,768 | 1 | 56.25 | 0.835 |
| Pusher displacement | DINO-WM | Mixed | 2,048 | 3 | 93.82 | 0.507 |
| Pusher displacement | DINO-WM | Mixed | 32,768 | 1 | 97.66 | 0.394 |
| Arm mass | LeWM | Mixed | 2,048 | 3 | 85.42 | 0.120 |
| Arm mass | LeWM | Mixed | 32,768 | 1 | 85.74 | 0.045 |
| Arm mass | PLDM | Mixed | 2,048 | 3 | 53.12 | 1.122 |
| Arm mass | PLDM | Mixed | 32,768 | 1 | 91.02 | 0.031 |
| Arm mass | DINO-WM | Mixed | 2,048 | 3 | 51.17 | 0.999 |
| Arm mass | DINO-WM | Mixed | 32,768 | 1 | 78.32 | 0.673 |
| Contact friction | LeWM | Mixed | 8,192 | 3 | 50.00 | 1.000 |
| Contact friction | LeWM | Mixed | 32,768 | 1 | 50.00 | 1.000 |

| Task | Model | Scheme | Pairs | $n$ | Accuracy (%) | NRE |
|---|---|---|---|---|---|---|
| Contact friction | PLDM | Mixed | 8,192 | 3 | 50.00 | 1.002 |
| Contact friction | PLDM | Mixed | 32,768 | 1 | 50.00 | 1.001 |
| Contact friction | DINO-WM | Mixed | 8,192 | 3 | 49.93 | 1.006 |
| Contact friction | DINO-WM | Mixed | 32,768 | 1 | 84.18 | 0.648 |
| Motion damping | LeWM | Mixed | 8,192 | 3 | 50.20 | 1.000 |
| Motion damping | LeWM | Mixed | 32,768 | 1 | 50.00 | 1.004 |
| Motion damping | PLDM | Mixed | 8,192 | 3 | 51.24 | 2.590 |
| Motion damping | PLDM | Mixed | 32,768 | 1 | 58.20 | 2.435 |
| Motion damping | DINO-WM | Mixed | 8,192 | 3 | 49.61 | 1.031 |
| Motion damping | DINO-WM | Mixed | 32,768 | 1 | 56.45 | 1.042 |
| Gripper carry rule | LeWM | Mixed | 2,048 | 3 | 50.00 | 1.000 |
| Gripper carry rule | LeWM | Mixed | 10,000 | 1 | 50.00 | 1.000 |
| Gripper carry rule | PLDM | Mixed | 2,048 | 3 | 50.00 | 1.003 |
| Gripper carry rule | PLDM | Mixed | 10,000 | 1 | 49.80 | 0.997 |
| Gripper carry rule | DINO-WM | Mixed | 2,048 | 3 | 50.78 | 0.996 |
| Gripper carry rule | DINO-WM | Mixed | 10,000 | 1 | 60.16 | 0.892 |
| Portal exit | LeWM | Mixed | 2,048 | 3 | 51.37 | 0.934 |
| Portal exit | LeWM | Mixed | 32,768 | 1 | 50.78 | 0.922 |
| Portal exit | PLDM | Mixed | 2,048 | 3 | 69.47 | 0.271 |
| Portal exit | PLDM | Mixed | 32,768 | 1 | 72.27 | 0.128 |
| Portal exit | DINO-WM | Mixed | 2,048 | 3 | 99.35 | 0.549 |
| Portal exit | DINO-WM | Mixed | 32,768 | 1 | 100.00 | 0.142 |

## H.2. Model Capacity

Both model sizes use the same 32,768-pair datasets, training seed, Mixed initialization, and 10-epoch recipe. The larger model changes the visual backbone from ViT-Tiny to ViT-Base, the representation width from 192 to 768, and predictor depth from six to nine layers. This is a joint expansion of the encoder and predictor, not a predictor-only intervention. Accuracy and NRE use Test; CEM measures planning success in the original environment.

| | | Accuracy (%) | | NRE | | CEM (%) | |
|---|---|---|---|---|---|---|---|
| Task | Model | 18M | 183M | 18M | 183M | 18M | 183M |
| Contact friction | LeWM | 50.00 | 50.00 | 1.000 | 1.000 | 76.67 | 74.00 |
| Contact friction | PLDM | 50.00 | 49.80 | 1.001 | 1.061 | 65.67 | 47.33 |
| Motion damping | LeWM | 50.00 | 50.20 | 1.004 | 1.003 | 78.00 | 71.33 |
| Motion damping | PLDM | 58.20 | 50.98 | 2.435 | 1.768 | 61.00 | 50.33 |

The four endpoint comparisons do not show a consistent improvement from increasing capacity under this recipe. These endpoint comparisons do not exclude gains from retuning the larger models.